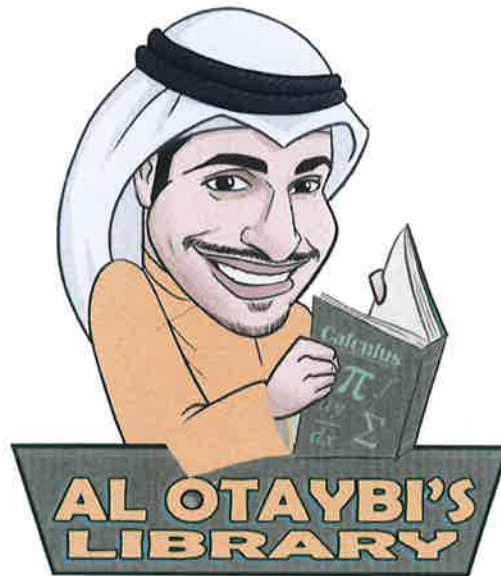


Kuwait University

2021/2022

Spring course



Calculus B

Final

Sections : 10.3 – 10.4 (part a/b)

10.3 Polar Coordinates.

10.4 Areas and Lengths in Polar Coordinates.

(17)

Eng. Mohammad Alotaybi 90048657 Twitter: @M_Alotaybi



10.3 – 10.4 Polar curves



10.3

$$r = \theta \dots$$

Convert a polar
equation to
 x and y equation

Arc length

Derivatives

Sketch

Area

طريقة وضع نقاط الـ Polar



- 1) أشوف الزاوية واعبر عنها بخط مستقيم.
- 2) أمشي على الخط بمقدار الطول.

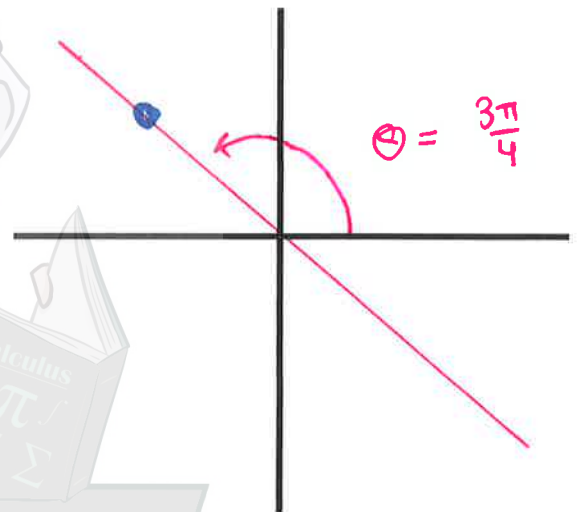
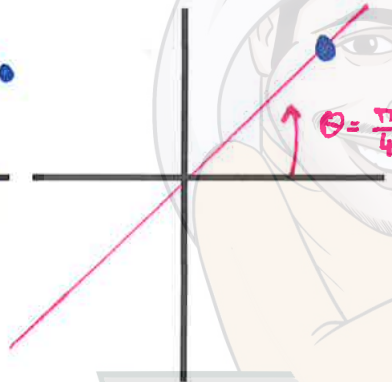
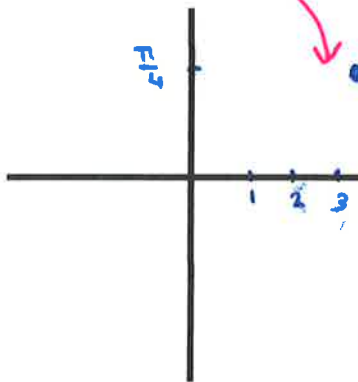


Plot each of the following polar points

$$P\left(3, \frac{\pi}{4}\right)$$

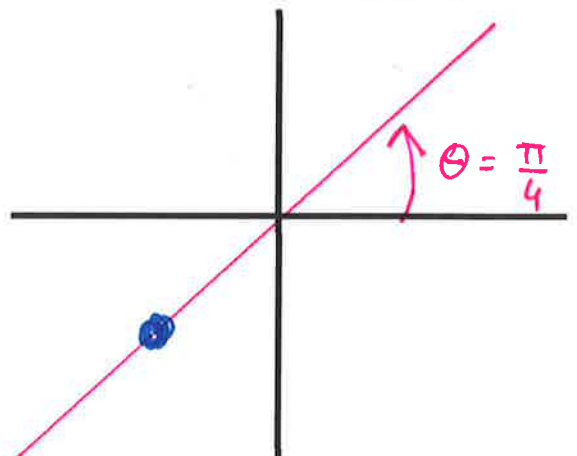
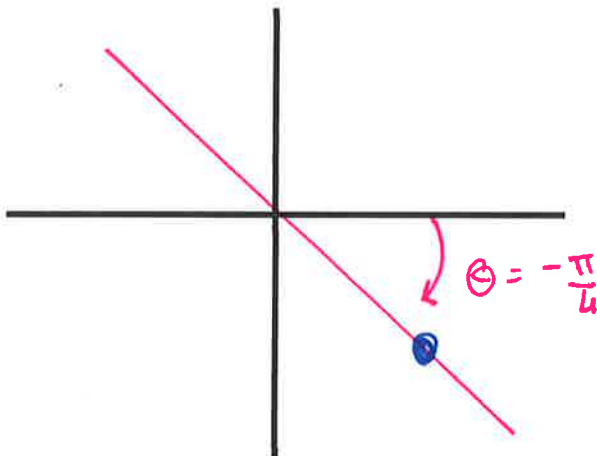
$$P\left(3, \frac{3\pi}{4}\right)$$

Big mistake!



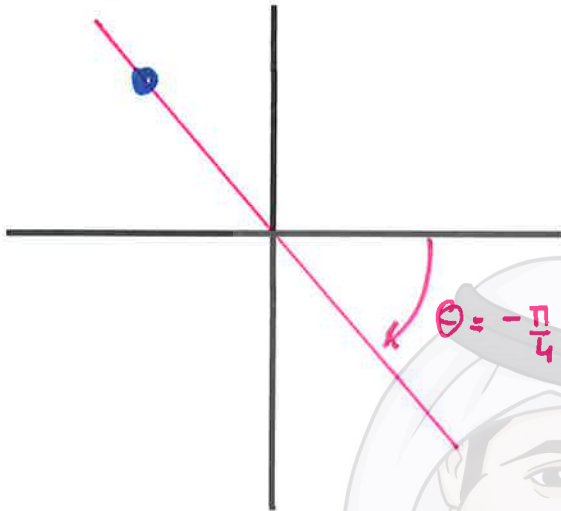
$$P\left(3, -\frac{\pi}{4}\right)$$

$$P\left(-3, \frac{\pi}{4}\right)$$

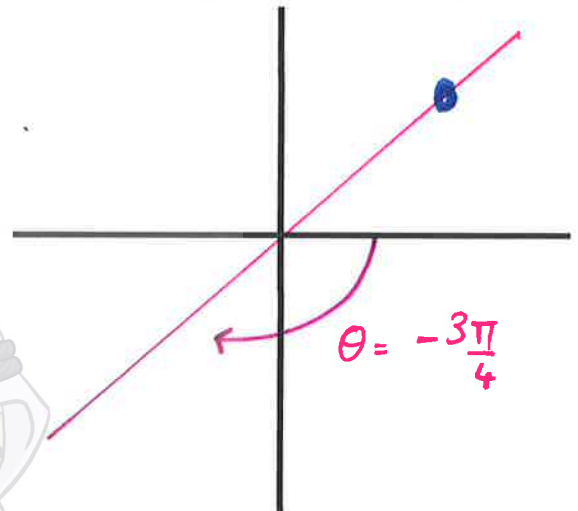


Plot each of the following polar points

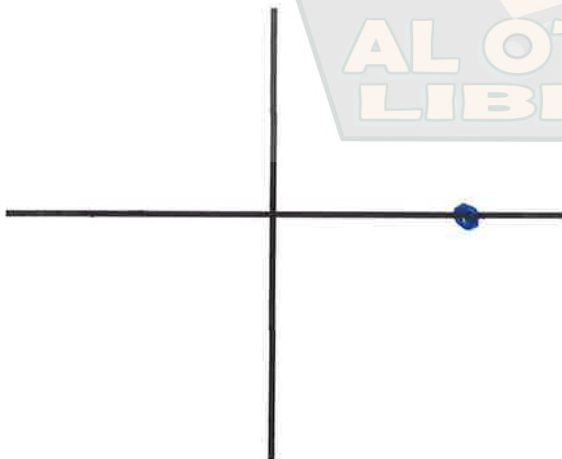
$$P\left(-3, -\frac{\pi}{4}\right)$$



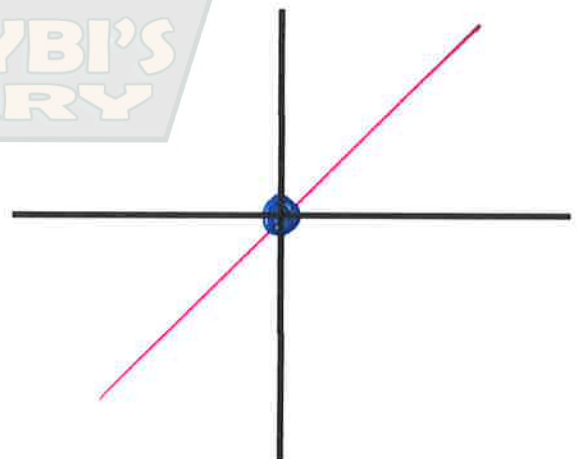
$$P\left(-3, -\frac{3\pi}{4}\right)$$



$$P(-3, \pi)$$



$$P\left(0, \frac{\pi}{4}\right)$$



Sketch the graph of the polar equation

$$r = 2 + 4\cos\theta$$



يتم اختيار أرقام عشوائية ولكن الأسهل

زاوية ناتجها حلو
و ممتماتها

الزوايا القائمة



$$r = 2 + 4\cos\theta$$

θ	0	$\frac{\pi}{3}$	$\frac{\pi}{2}$	$\frac{2\pi}{3}$	π	$\frac{4\pi}{3}$	$\frac{3\pi}{2}$	$\frac{5\pi}{3}$	2π
r	6	4	2	0	-2	0	2	4	6

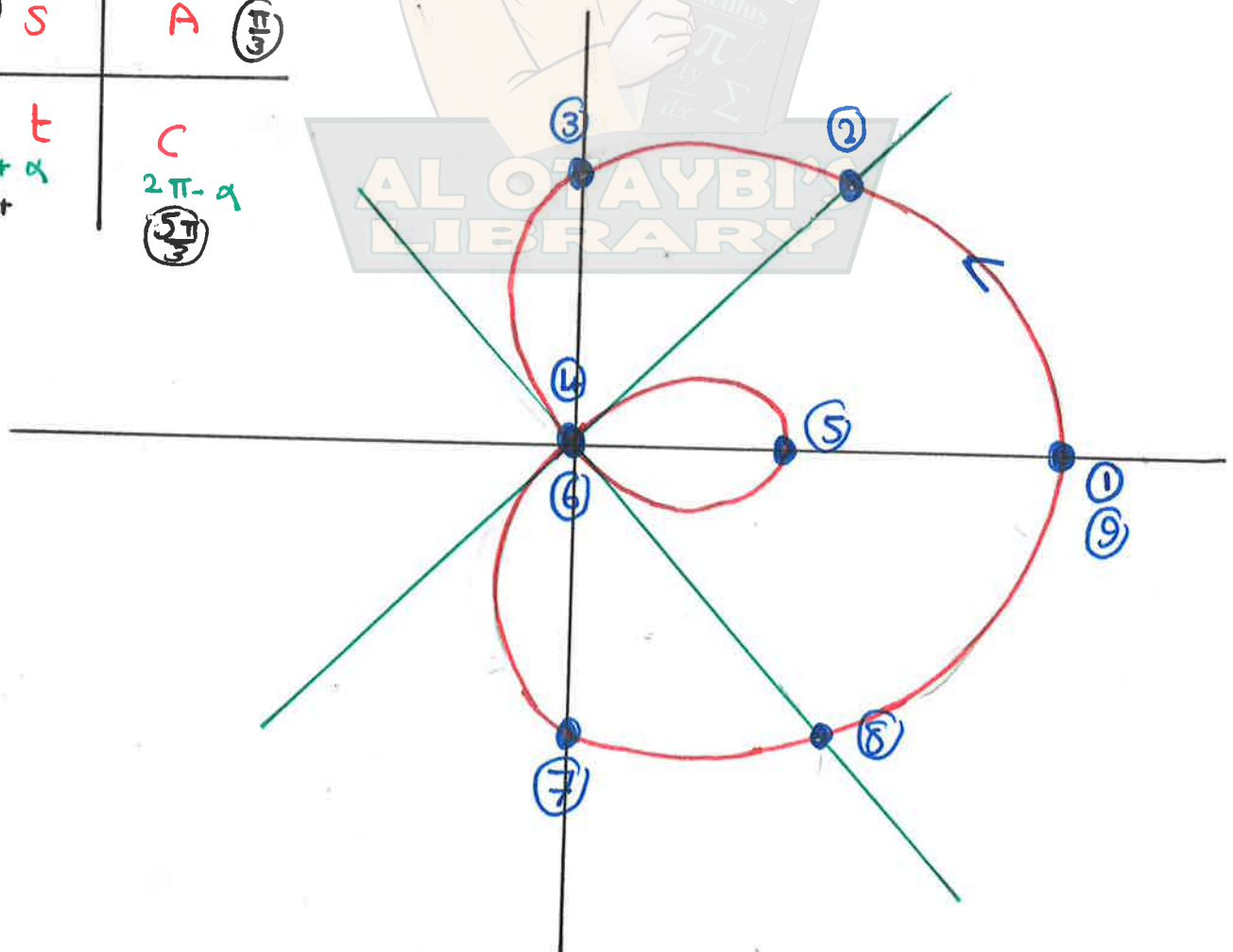
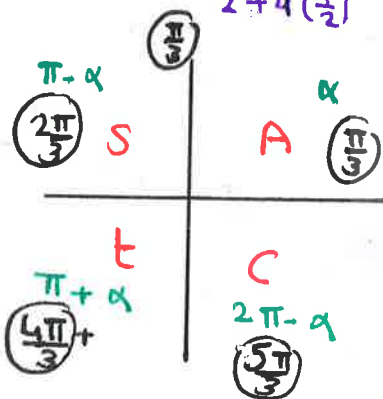
$$2 + 4\left(\frac{1}{2}\right)$$

$$2 + 4\left(-\frac{1}{2}\right)$$

$$2 + 4(-1)$$

$$2 + 4\left(-\frac{1}{2}\right)$$

$$2 + 4\left(\frac{1}{2}\right)$$



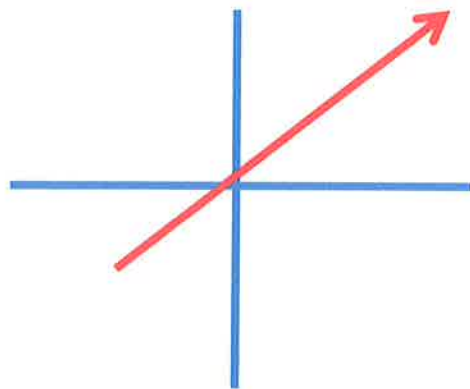


Graphs

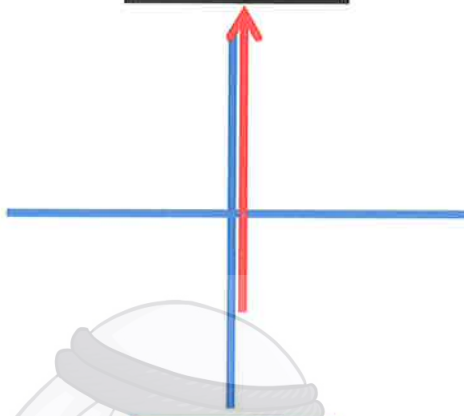


Line

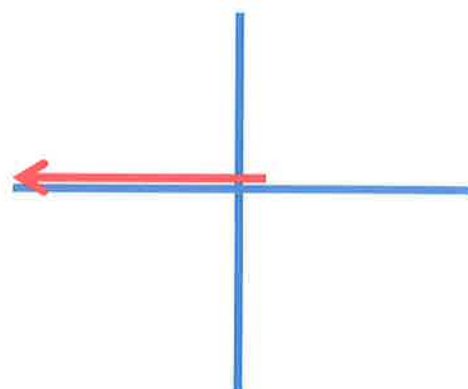
$$\theta = \frac{\pi}{4}$$



$$\theta = \frac{\pi}{2}$$

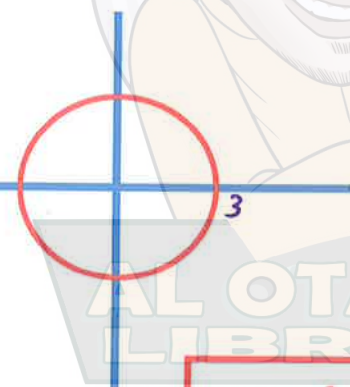


$$\theta = \pi$$

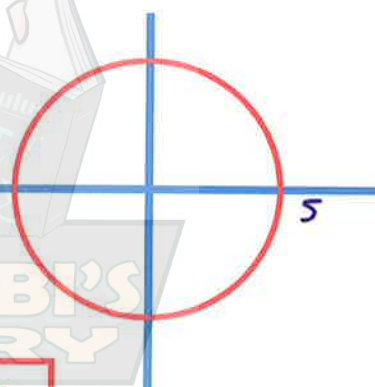


Circle

$$r = 3$$

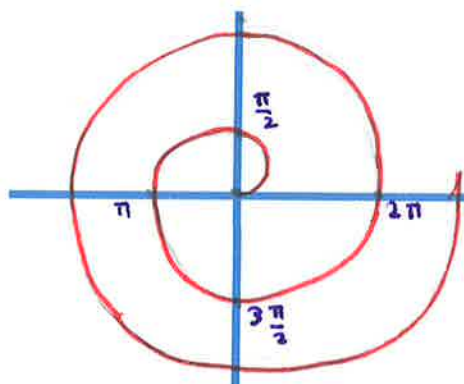


$$r = 5$$



Spiral of arch

$$r = \theta$$

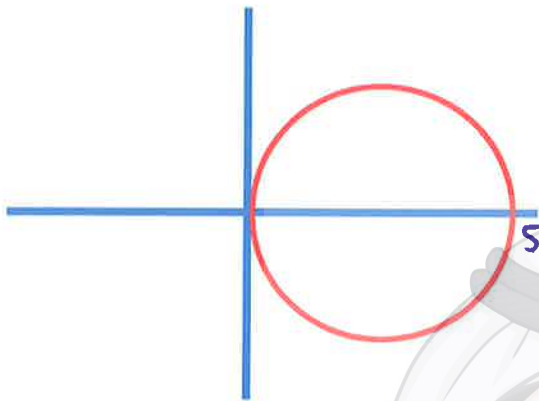




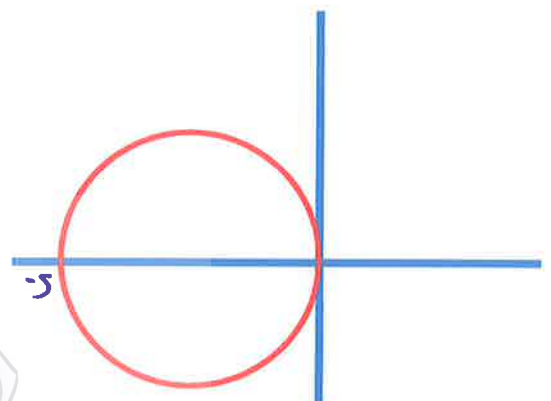
Circles



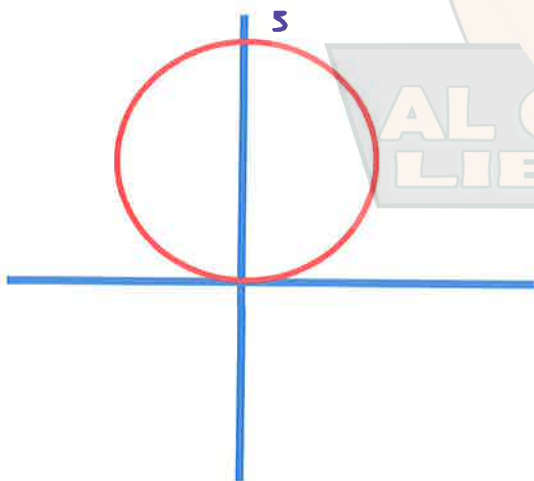
$$r = 5\cos\theta$$



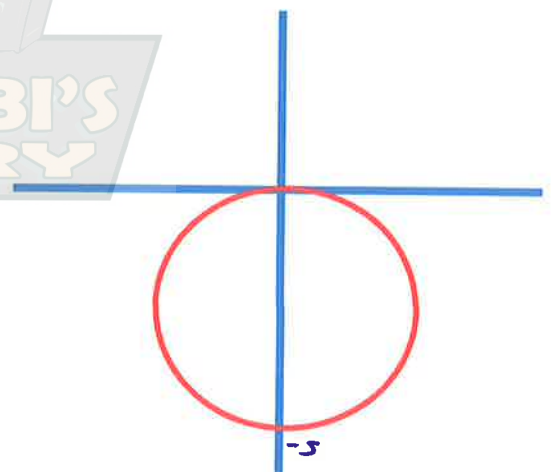
$$r = -5\cos\theta$$



$$r = 5\sin\theta$$



$$r = -5\sin\theta$$



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LIBRARY

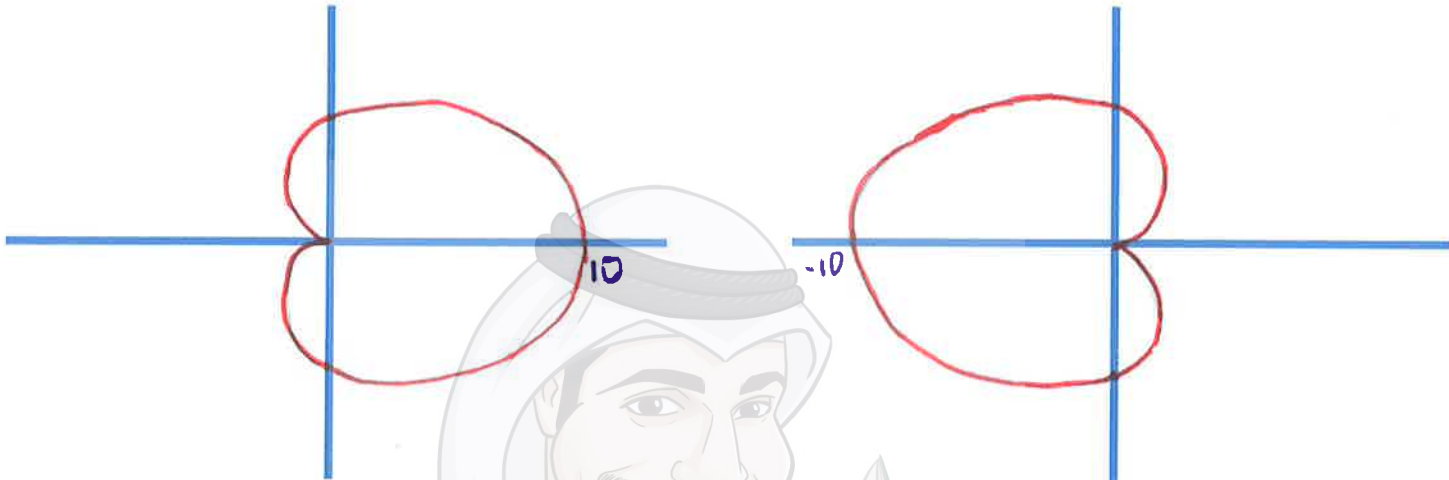


Cardioid



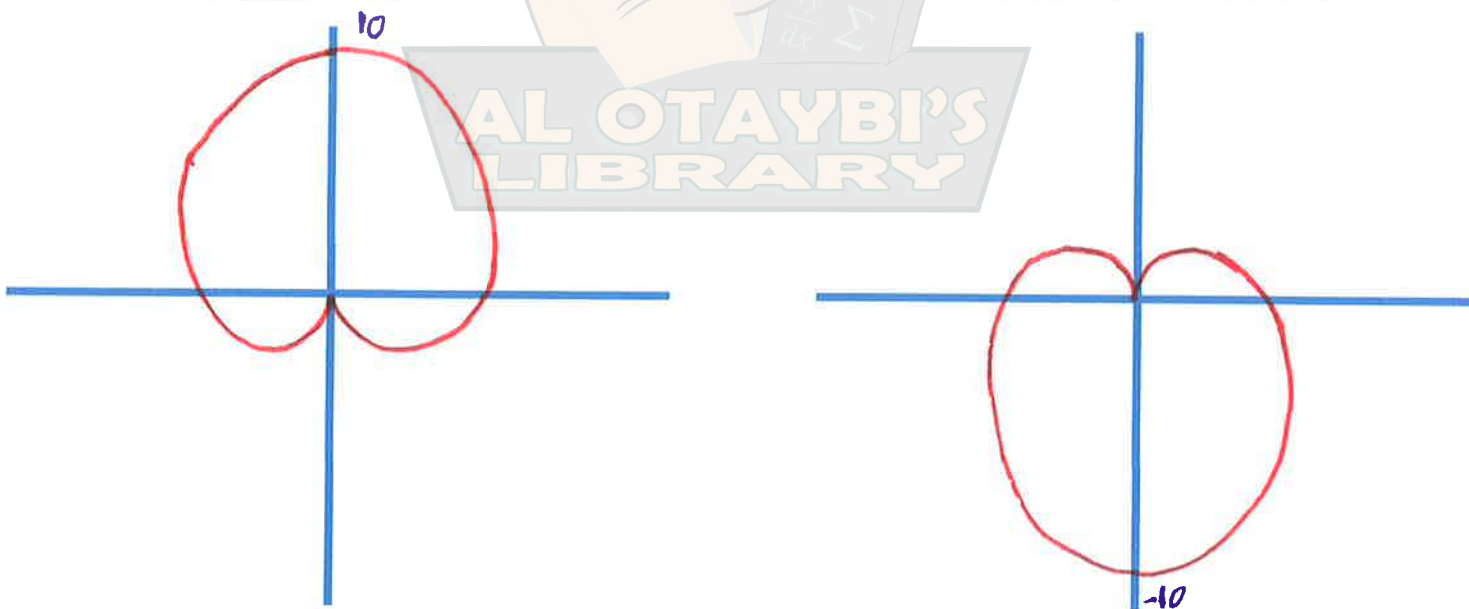
$$r = 5 + 5\cos\theta$$

$$r = 5 - 5\cos\theta$$



$$r = 5 + 5\sin\theta$$

$$r = 5 - 5\sin\theta$$

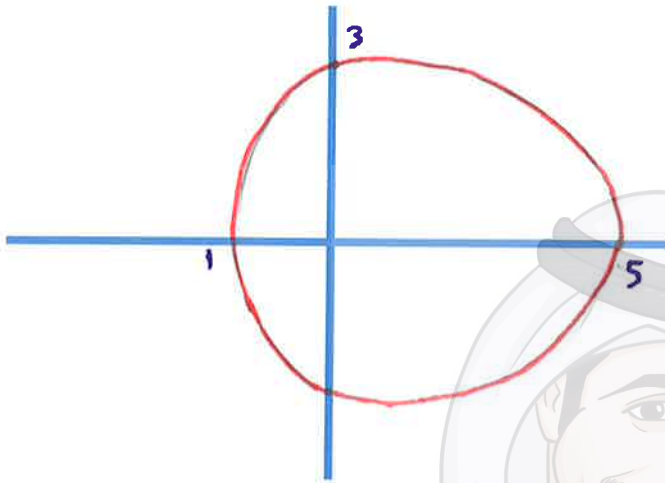




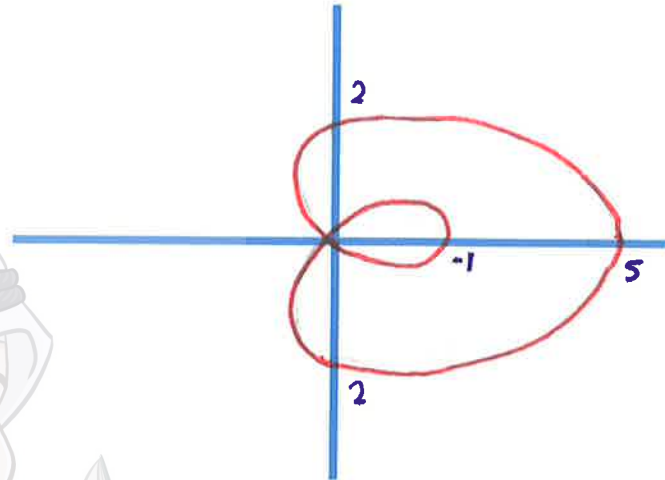
Limacons



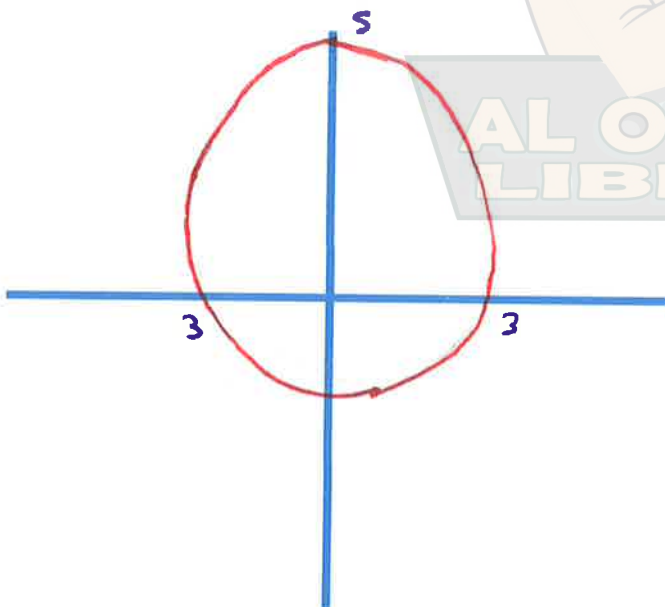
$$r = 3 + 2\cos\theta$$



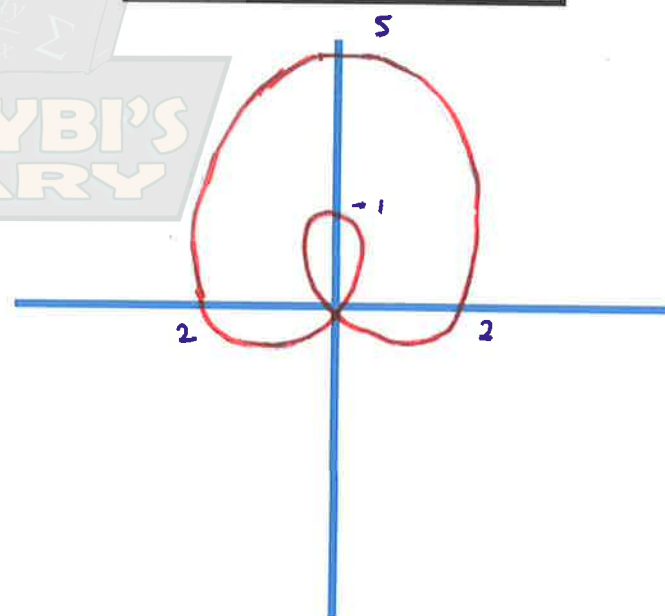
$$r = 2 + 3\cos\theta$$



$$r = 3 + 2\sin\theta$$



$$r = 2 + 3\sin\theta$$

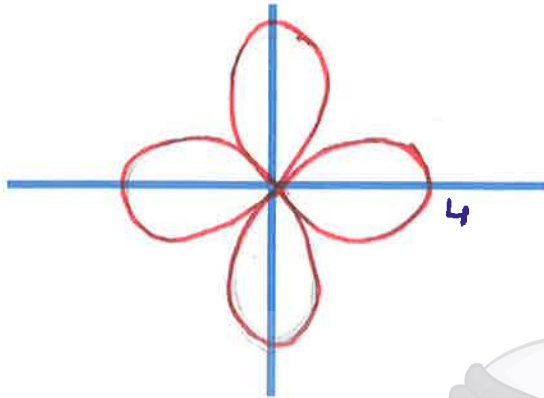




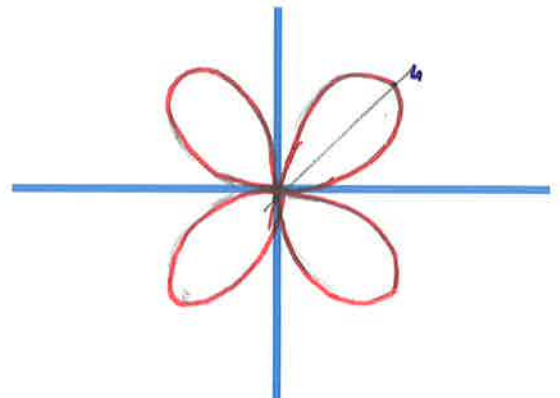
Leaved Roses



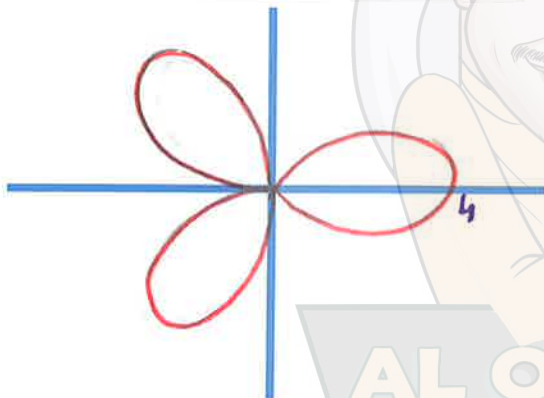
$$r = 4\cos(2\theta)$$



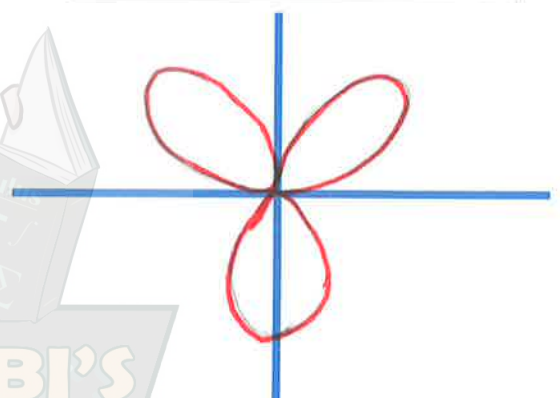
$$r = 4\sin(2\theta)$$



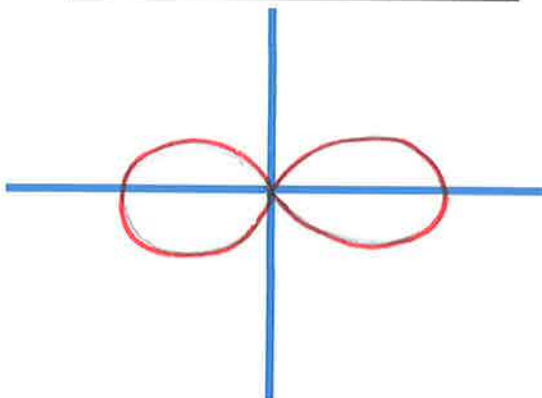
$$r = 4\cos(3\theta)$$



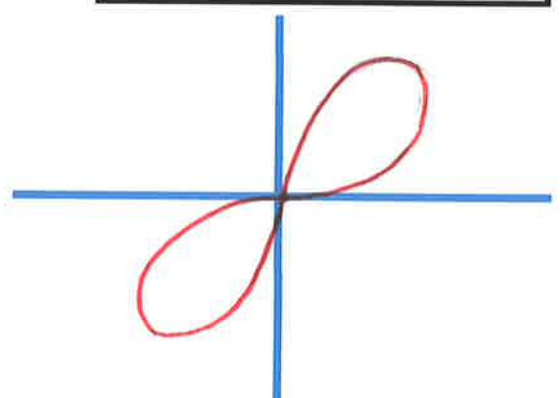
$$r = 4\sin(3\theta)$$



$$r^2 = 4\cos(2\theta)$$
$$r = 2\sqrt{\cos(2\theta)}$$



$$r^2 = 4\sin(2\theta)$$
$$r = 2\sqrt{\sin(2\theta)}$$



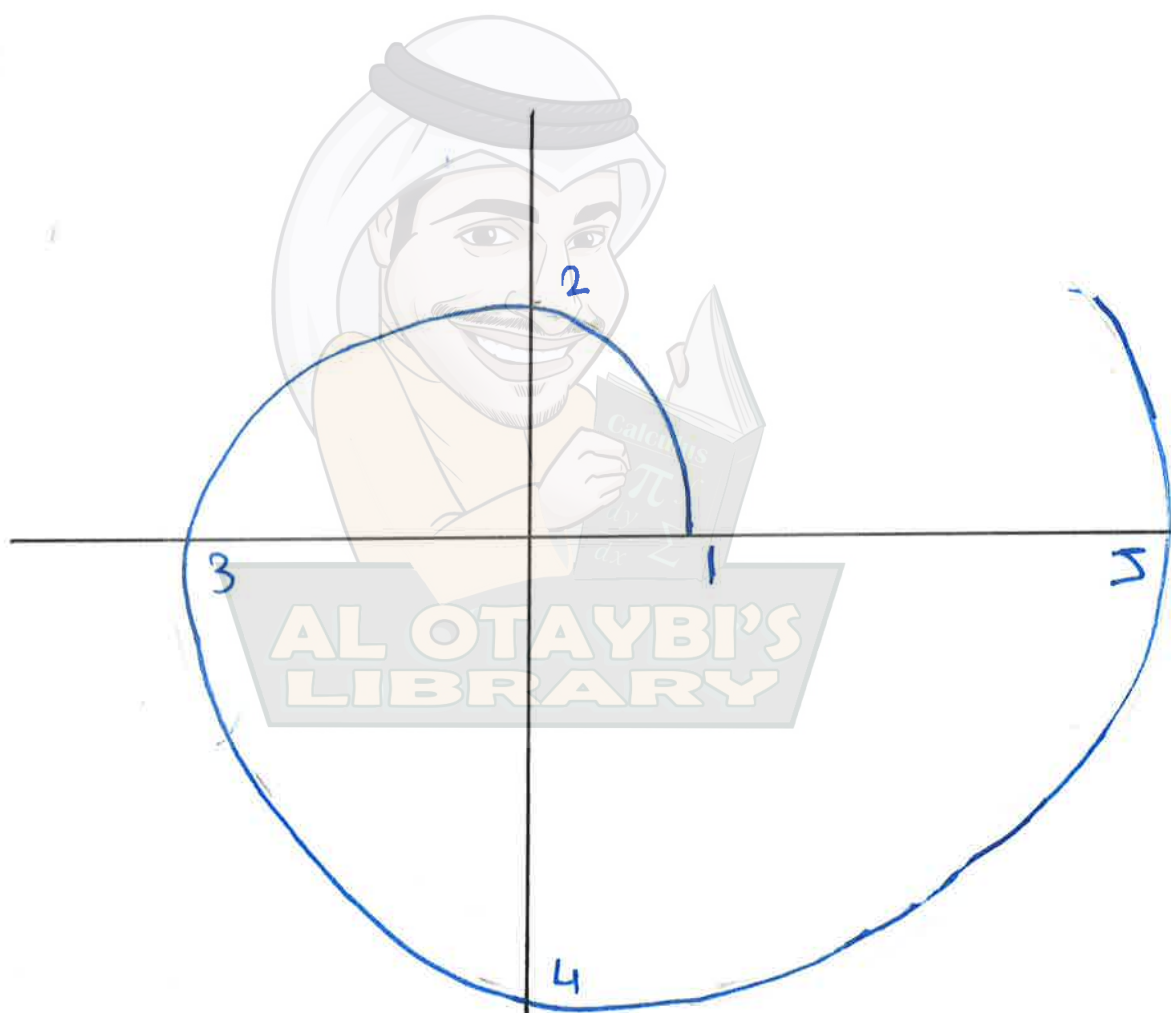
AL OTAYBI'S
LIBRARY

Sketch the graph of the polar equation

$$r = \frac{2}{\pi}\theta + 1$$

θ	0	$\frac{\pi}{2}$	π	$\frac{3\pi}{2}$	2π
r	1	2	3	4	5

$$\frac{2}{\pi} \frac{\pi}{2} + 1$$





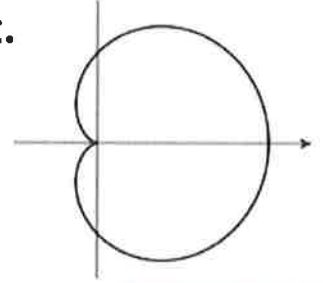
Multiple choice



Select the polar equation that matches the given plot.

(A) $r = 1 - \sin\theta$ (B) $r = 1 + \sin\theta$

(C) $r = 1 - \cos\theta$ (D) $r = 1 + \cos\theta$



May 2017

(D) ✓





Polar coordinates



هو إحداثي متعلق بزاوية وطول

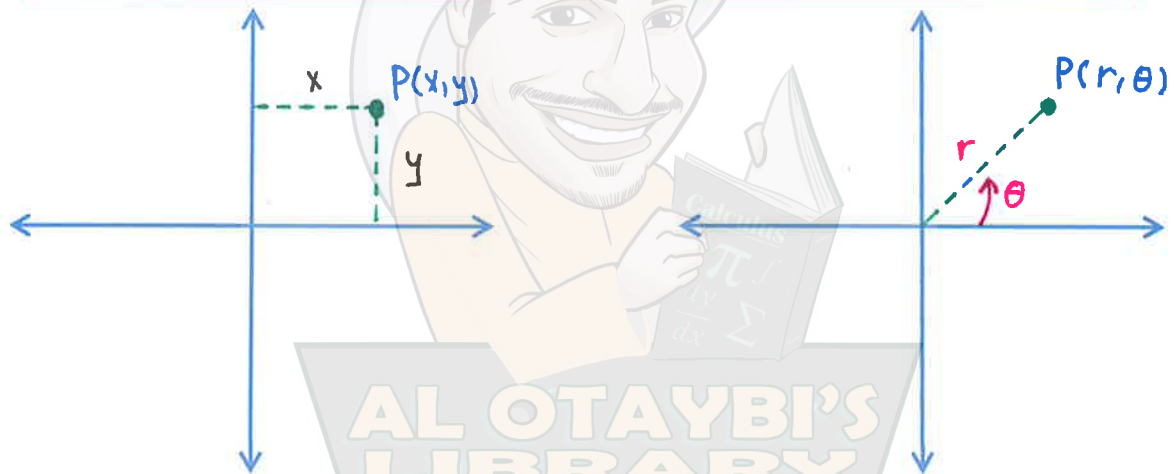
$$P(r, \theta)$$

طول

زاوية

Cartesian coordinate
rectangle coordinate

Polar coordinate



تحويل من r, θ

الى
 x, y

$$x = r \cos \theta$$

$$y = r \sin \theta$$

تحويل من x, y

الى
 r, θ

$$r^2 = x^2 + y^2$$

$$\tan \theta = \frac{y}{x}$$

The point $P(\sqrt{3}, 1)$ is represented on the Cartesian (rectangular system).

Represent it in the polar coordinate. $(0 \leq \theta \leq \frac{\pi}{2}) (r \geq 0)$

$$\Rightarrow r^2 = x^2 + y^2$$



$$r^2 = x^2 + y^2$$

$$\tan \theta = \frac{y}{x}$$



$$r^2 = (\sqrt{3})^2 + (1)^2$$

$$r^2 = 3 + 1$$

$$r^2 = 4$$

$$r = 2$$

$\in \text{Domain}$

$$r = -2$$

$\notin \text{Domain}$

$$\Rightarrow \tan \theta = \frac{y}{x}$$

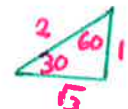
$$\tan \theta = \frac{1}{\sqrt{3}}$$

$$\theta = \frac{\pi}{6}$$

$\in \text{Domain}$

$$\theta = \pi + \frac{\pi}{6} = \frac{7\pi}{6}$$

$\notin \text{Domain}$

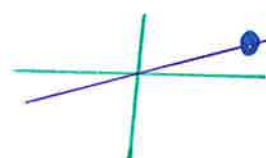


$$\Rightarrow \therefore P(r, \theta)$$

$$= P(2, \frac{\pi}{6})$$



حل أو متغير



The point $P(1, 1)$ is represented on the Cartesian (rectangular system).

Represent it in the polar coordinate. $(0 \leq \theta \leq \frac{\pi}{2})$ ($r \geq 0$)

$$\Rightarrow r^2 = x^2 + y^2$$

$$r^2 = (1)^2 + (1)^2$$

$$r^2 = 1 + 1$$

$$r^2 = 2$$

$$r = \sqrt{2} \in \text{Domain}$$

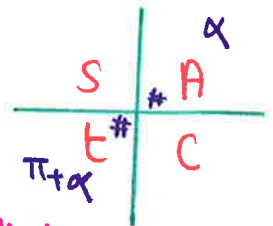
$$r = -\sqrt{2} \notin \text{Domain}$$

$$\Rightarrow \tan \theta = \frac{y}{x}$$

$$\tan \theta = \frac{1}{1}$$

$$\theta = \frac{\pi}{4} \in \text{Domain}$$

$$\theta = \pi + \frac{\pi}{4} = \frac{5\pi}{4} \notin \text{Domain}$$



$$\Rightarrow \therefore P(r, \theta) = P(\sqrt{2}, \frac{\pi}{4})$$

Change the polar curve $r^2 \sin(2\theta) = 2$
to a rectangular system.

$$r^2 \sin(2\theta) = 2$$

$$r^2 2 \sin\theta \cos\theta = 2$$

$$2 r^2 \sin\theta \cos\theta = 2$$

$$2 r \cdot r \sin\theta \cos\theta = 2$$

$$2 r \sin\theta r \cos\theta = 2$$

$$\begin{array}{l} 2 y x = 2 \\ \div 2 \quad \xrightarrow{\text{للبسيط}} \\ y x = 1 \end{array}$$



للتحويل من

r, θ



الى

x, y

$$x = r \cos\theta$$

$$y = r \sin\theta$$

Find the Cartesian equation for the polar curve

$$r^2 = \csc(2\theta)$$

$$r^2 = \frac{1}{\sin(2\theta)}$$

$$r^2 \sin(2\theta) = 1$$

$$r^2 2 \sin\theta \cos\theta = 1$$

$$2 r^2 \sin\theta \cos\theta = 1$$

$$2 r \cdot r \sin\theta \cos\theta = 1$$

$$2 r \sin\theta r \cos\theta = 1$$

$$2 y x = 1$$

Show that the polar equation represent a vertical line in xy-plane

$$r = \frac{3}{4} \sec \theta$$

$$r = \frac{3}{4} \cdot \frac{1}{\cos \theta}$$

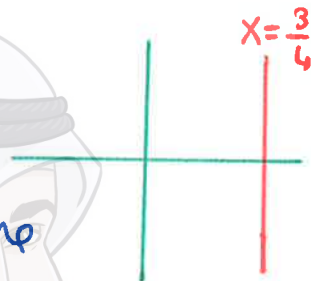
* $\cos \theta$

$$r \cos \theta = \frac{3}{4}$$



$$x = \frac{3}{4}$$

which is vertical line



للتحويل من

r, θ



الى

x, y

$$x = r \cos \theta$$

$$y = r \sin \theta$$

Show that the polar equation represent a horizontal line

$$r = 3 \csc \theta$$

$$r = 3 \cdot \frac{1}{\sin \theta}$$

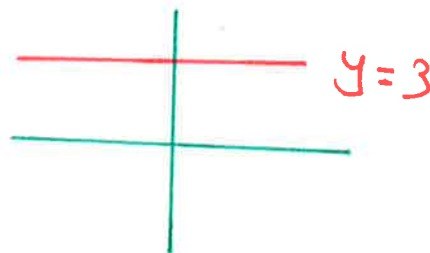
* $\sin \theta$

$$r \sin \theta = 3$$



$$y = 3$$

which is horizontal line



Show that the polar equation

$r = 2\sin\theta + 4\cos\theta$ represent a circle ,

then find its center and radius.

$$r = 2\sin\theta + 4\cos\theta$$

* r

$$r^2 = 2r\sin\theta + 4r\cos\theta$$

$$x^2 + y^2 = 2y + 4x$$

$$x^2 - 4x + y^2 - 2y = 0$$

$$(x - 2)^2 - 4 + (y - 1)^2 - 1 = 0$$

$$(x - 2)^2 + (y - 1)^2 - 5 = 0$$

$$(x - 2)^2 + (y - 1)^2 = 5 \Rightarrow \text{Equation of circle}$$

$$\text{Center} = (2, 1)$$

$$\text{Radius} = \sqrt{5}$$

يبيني أثبت انها معادلة دائرة
لازم أوصلها الى شكل معادلة الدائرة

$$(x - a)^2 + (y - b)^2 = c$$

لازم أحولها الى معادلة
 x, y

$$x = r\cos\theta$$

$$y = r\sin\theta$$

Show that the polar equation

$r = a \sin \theta + b \cos \theta$ represent a circle ,
then find its center and radius.

$$r = a \sin \theta + b \cos \theta$$

*r

$$r^2 = a r \sin \theta + b r \cos \theta$$

$$x^2 + y^2 = ay + bx$$

$$x^2 - bx + y^2 - ay = 0$$

$$\left(x - \frac{b}{2}\right)^2 - \frac{b^2}{4} + \left(y - \frac{a}{2}\right)^2 - \frac{a^2}{4} = 0$$

$$\left(x - \frac{b}{2}\right)^2 + \left(y - \frac{a}{2}\right)^2 = \frac{a^2}{4} + \frac{b^2}{4}$$

$$\left(x - \frac{b}{2}\right)^2 + \left(y - \frac{a}{2}\right)^2 = \frac{a^2 + b^2}{4} \rightarrow \text{Equation of Circle}$$

$$\text{Center} = \left(\frac{b}{2}, \frac{a}{2}\right)$$

$$\text{Radius} = \sqrt{\frac{a^2 + b^2}{4}} = \frac{\sqrt{a^2 + b^2}}{2} = \cancel{\frac{a+b}{2}}$$

Find a polar equation $r = f(\theta)$ for the curve represented by Cartesian equation

$$x^2 + y^2 = 10x + 6y$$

Fall 2017

$$r^2 = 10r\cos\theta + 6r\sin\theta$$

(if $r \neq 0$) $\div r$

$$r = 10\cos\theta + 6\sin\theta$$

Let $C_1: r = 2\cos\theta$ and $C_2: r = \sqrt{3}$ be polar curves.

Write the Cartesian equations for C_1 and C_2 .

Fall 2017

$$r = 2\cos\theta$$

*r

$$r^2 = 2r\cos\theta$$

$$x^2 + y^2 = 2x$$

$$x^2 - 2x + y^2 = 0$$

$$(x-1)^2 - 1 + y^2 = 0 \rightarrow (x-1)^2 + y^2 = 1$$

$$r = \sqrt{3}$$

Square both sides

$$r^2 = 3$$

$$x^2 + y^2 = 3$$



Multiple choice



The point with polar coordinates $(2\sqrt{2}, -\frac{\pi}{4})$ has Cartesian coordinates $(x, y) =$

Jan 2017

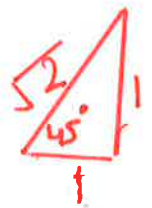
- (a) $(2\sqrt{2}, -2\sqrt{2})$ (b) $(\sqrt{6}, -\sqrt{2})$ (c) $(2, 2)$ (d) $(2, -2)$ (e) $(-2, 2)$

$$(r, \theta) = (2\sqrt{2}, -\frac{\pi}{4})$$

r θ

$$x = r \cos \theta$$

$$x = 2\sqrt{2} \cos(-\frac{\pi}{4}) = 2\sqrt{2} \cos(\frac{\pi}{4})$$
$$= 2\sqrt{2} \cdot \frac{1}{\sqrt{2}} = \boxed{2}$$



$$y = r \sin \theta$$

$$y = 2\sqrt{2} \sin(-\frac{\pi}{4}) = -2\sqrt{2} \sin(\frac{\pi}{4})$$
$$= -2\sqrt{2} \cdot \frac{1}{\sqrt{2}} = \boxed{-2}$$

$$\Rightarrow \therefore P(x, y) = P(2, -2) \quad (d)$$



Multiple choice



The polar curve $r = 2\cos\theta + 3\sin\theta$ represents a circle centered at

- (a) $(1, -\frac{3}{2})$ (b) $(-1, -\frac{3}{2})$ (c) $(-1, \frac{3}{2})$ (d) $(1, \frac{3}{2})$ (e) $(\frac{3}{2}, 1)$

Fall 2018

$$r = 2\cos\theta + 3\sin\theta$$

* r

$$r^2 = 2r\cos\theta + 3r\sin\theta$$

$$x^2 + y^2 = 2x + 3y$$

$$x^2 - 2x + y^2 - 3y = 0$$

$$(x-1)^2 - 1 + (y-\frac{3}{2})^2 - \frac{9}{4} = 0$$

$$(x-1)^2 + (y-\frac{3}{2})^2 = \frac{9}{4} + 1$$

$$\text{Center} = (1, \frac{3}{2}) \quad (d)$$



Polar curves



Arc length

$$L = \int_a^b \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2} d\theta$$

عطائي حدود التكامل



تطبيق مباشر للقانون

ما عطائي حدود التكامل



(1) الرسم
(2) بحث السيمتري
(3) ايجاد حدود التكامل

Find the arc length of the polar curve

$$r = \theta^2, \quad 0 \leq \theta \leq \pi$$

$$L = \int_0^{\pi} \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2} d\theta$$

$$r = \theta^2 \longrightarrow r^2 = (\theta^2)^2 = \theta^4$$

$$\frac{dr}{d\theta} = 2\theta \longrightarrow \left(\frac{dr}{d\theta}\right)^2 = (2\theta)^2 = 4\theta^2$$

$$\implies r^2 + \left(\frac{dr}{d\theta}\right)^2 = \theta^4 + 4\theta^2 = \theta^2(\theta^2 + 4)$$

$$L = \int_0^{\pi} \sqrt{\theta^2(\theta^2 + 4)} d\theta$$

$$= \int_0^{\pi} |\theta| \sqrt{\theta^2 + 4} d\theta$$

$\hookrightarrow \theta = 0 \notin (0, \pi)$

$$= \int_0^{\pi} \theta (\theta^2 + 4)^{\frac{1}{2}} d\theta$$

Let $u = \theta^2 + 4$

$$du = 2\theta d\theta \longrightarrow \frac{1}{2} du = \theta d\theta$$

$$= \frac{1}{2} \int u^{\frac{1}{2}} du = \frac{1}{2} \cdot \frac{u^{\frac{3}{2}}}{\frac{3}{2}}$$

$$= \frac{1}{3} \left[(\theta^2 + 4)^{\frac{3}{2}} \right]_0^{\pi}$$

$$= \frac{1}{3} \left[(\pi^2 + 4)^{\frac{3}{2}} - (4)^{\frac{3}{2}} \right]$$

Find the arc length of the polar curve

$$r = \theta, \quad 0 \leq \theta \leq 3$$

$$L = \int_0^3 \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2} d\theta$$

$$r = \theta \rightarrow r^2 = \theta^2$$

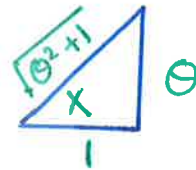
$$\frac{dr}{d\theta} = 1 \rightarrow \left(\frac{dr}{d\theta}\right)^2 = 1$$

$$\Rightarrow r^2 + \left(\frac{dr}{d\theta}\right)^2 = \theta^2 + 1$$

$$L = \int_0^3 \sqrt{\theta^2 + 1} d\theta$$

$$\text{Let } \theta = \tan x$$

$$d\theta = \sec^2 x dx$$



$$= \int \sqrt{\tan^2 x + 1} \sec^2 x dx$$

$$\rightarrow \sqrt{\sec^2 x} = \sec x$$

$$= \int \sec x \sec^2 x dx$$

$$= \int \sec^3 x dx$$

$\Rightarrow$
See Next page
Plz!

1) By parts

2) هنا

3) الجزء
"لولي"

$$\int \sec^3 x \, dx = I$$

$$\int \sec x \sec^2 x \, dx$$

$$u = \sec x$$

$$dv = \sec^2 x \, dx$$

$$du = \sec x \tan x \, dx \quad v = \tan x$$

$$\sec x \tan x - \int \sec x \tan^2 x \, dx$$

$$\int \sec x (\sec^2 x - 1) \, dx$$

$$\int (\sec^3 x - \sec x) \, dx$$

$$I - \ln |\sec x + \tan x|$$

$$I = \sec x \tan x - [I - \ln |\sec x + \tan x|]$$

$$I = \sec x \tan x - I + \ln |\sec x + \tan x|$$

$$2I = \sec x \tan x + \ln |\sec x + \tan x|$$

$$I = \frac{1}{2} [\sec x \tan x + \ln |\sec x + \tan x|]$$

$$= \frac{1}{2} \left[\frac{\sqrt{\theta^2 + 1}}{1} \cdot \frac{\theta}{1} + \ln \left| \frac{\sqrt{\theta^2 + 1}}{1} + \frac{\theta}{1} \right| \right]_0^3$$

$$= \frac{1}{2} \left[\left(\frac{\sqrt{10}}{1} (3) + \ln \left| \frac{\sqrt{10}}{1} + 3 \right| \right) - (0 + \ln 1) \right]$$

$\downarrow 0$

Find the arc length of the polar curve

$$r = \cos^2\left(\frac{\theta}{2}\right) \quad \forall \quad \theta \in [0, 2\pi]$$

$$L = \int_0^{2\pi} \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2} d\theta$$

$$r = \cos^2\left(\frac{\theta}{2}\right) \longrightarrow r^2 = \cos^4\left(\frac{\theta}{2}\right)$$

$$\frac{dr}{d\theta} = 2\cos\left(\frac{\theta}{2}\right) \left[-\sin\left(\frac{\theta}{2}\right) \cdot \frac{1}{2}\right]$$

$$= -\cos\left(\frac{\theta}{2}\right) \sin\left(\frac{\theta}{2}\right) \longrightarrow r^2 = \left[-\cos\left(\frac{\theta}{2}\right) \sin\left(\frac{\theta}{2}\right)\right]^2$$

$$= \cos^2\left(\frac{\theta}{2}\right) \sin^2\left(\frac{\theta}{2}\right)$$

$$\Rightarrow r^2 + \left(\frac{dr}{d\theta}\right)^2 = \cos^4\left(\frac{\theta}{2}\right) + \cos^2\left(\frac{\theta}{2}\right) \sin^2\left(\frac{\theta}{2}\right)$$

$$= \cos^2\left(\frac{\theta}{2}\right) \left[\cos^2\left(\frac{\theta}{2}\right) + \sin^2\left(\frac{\theta}{2}\right)\right]$$

$$= \cos^2\left(\frac{\theta}{2}\right) (1) = \cos^2\left(\frac{\theta}{2}\right)$$

$$L = \int_0^{2\pi} \sqrt{\cos^2\left(\frac{\theta}{2}\right)} d\theta = \int_0^{2\pi} \left|\cos\frac{\theta}{2}\right| d\theta$$

$\begin{array}{c|c} S & A \\ \hline E & C \end{array}$
 $\frac{\theta}{2} = \frac{\pi}{2}$
 $\theta = \pi$
 $\in [0, 2\pi]$

$$= \int_0^{\pi} \cos\frac{\theta}{2} d\theta - \int_{\pi}^{2\pi} \cos\frac{\theta}{2} d\theta$$

$$= \left[\frac{\sin\frac{\theta}{2}}{\frac{1}{2}}\right]_0^{\pi} - \left[\frac{\sin\frac{\theta}{2}}{\frac{1}{2}}\right]_{\pi}^{2\pi}$$

$$= 2 \left[\sin\frac{\pi}{2} - 0\right] - 2 \left[\sin\pi - \sin\frac{\pi}{2}\right]$$

$$= 2(1) - 2(0 - 1) = 2 + 2 = 4$$



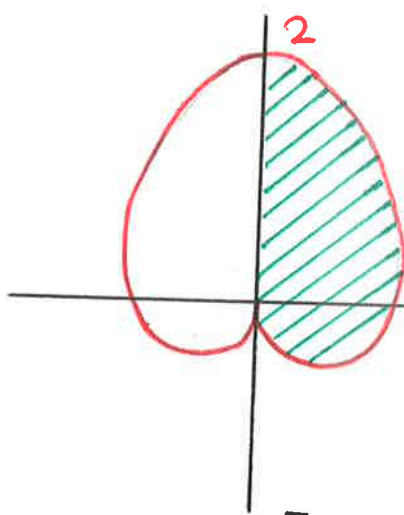
Tricky question !!



Find the length of the cardioid

$$r = 1 + \sin\theta$$

$$L = \int \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2} d\theta$$



$$2 = 1 + \sin\theta$$

$$\sin\theta = 1$$

$$\theta = \frac{\pi}{2}$$

$$0 = 1 + \sin\theta$$

$$\sin\theta = -1$$

$$\theta = \frac{3\pi}{2}$$

$$\text{or } \theta = -\frac{\pi}{2}$$

Symmetry

$$L = 2 \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2} d\theta$$

$$r = 1 + \sin\theta \rightarrow r^2 = (1 + \sin\theta)^2 = 1 + 2\sin\theta + \sin^2\theta$$

$$\frac{dr}{d\theta} = \cos\theta \rightarrow \left(\frac{dr}{d\theta}\right)^2 = \cos^2\theta$$

$$\Rightarrow r^2 + \left(\frac{dr}{d\theta}\right)^2 = 1 + 2\sin\theta + \sin^2\theta + \cos^2\theta$$

$$= 1 + 2\sin\theta + 1 = 2 + 2\sin\theta$$

$$= 2(1 + \sin\theta)$$

$$L = 2 \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sqrt{2(1+\sin\theta)} d\theta$$

$$= 2\sqrt{2} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sqrt{1+\sin\theta} d\theta * \frac{\sqrt{1-\sin\theta}}{\sqrt{1-\sin\theta}}$$

$$= 2\sqrt{2} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\sqrt{1-\sin^2\theta}}{\sqrt{1-\sin\theta}} d\theta$$

$\cos^2\theta + \sin^2\theta = 1$
 $\cos^2\theta = 1 - \sin^2\theta$

$$= 2\sqrt{2} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\sqrt{\cos^2\theta}}{\sqrt{1-\sin\theta}} d\theta \rightarrow |\cos\theta|$$

هناك خرج $\frac{\pi}{2}$

$$= 2\sqrt{2} \lim_{t \rightarrow \frac{\pi}{2}^-} \int_{-\frac{\pi}{2}}^t \cos\theta (1-\sin\theta)^{-\frac{1}{2}} d\theta$$

$$2\sqrt{2} \lim_{t \rightarrow \frac{\pi}{2}^-} \left[-2\sqrt{1-\sin\theta} \right]_{-\frac{\pi}{2}}^t$$

$$-4\sqrt{2} \lim_{t \rightarrow \frac{\pi}{2}^-} \left[\sqrt{1-\sin t} - \sqrt{1-(-1)} \right]$$

$$-4\sqrt{2} [0 - \sqrt{2}]$$

$$4(2) = \underline{\underline{8}}$$

$$I_1 = \int \cos\theta (1-\sin\theta)^{-\frac{1}{2}} d\theta$$

$$u = 1 - \sin\theta$$

$$du = -\cos\theta d\theta$$

$$-du = \cos\theta d\theta$$

$$-\int u^{-\frac{1}{2}} du = -\frac{u^{\frac{1}{2}}}{\frac{1}{2}} + c$$

$$= -2\sqrt{u} + c$$

$$= -2\sqrt{1-\sin\theta} + c$$



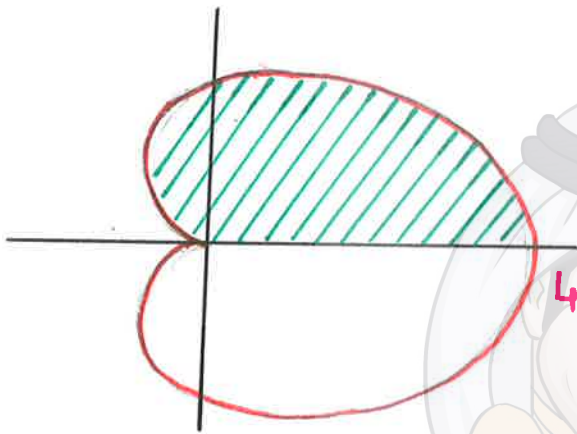
Tricky question !!



Find the length of the curve

$$r = 4\cos^2\left(\frac{\theta}{2}\right)$$

$$L = \int \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2} d\theta$$



$$r = 4\cos^2\left(\frac{\theta}{2}\right)$$

$$r = 4 \cdot \frac{1}{2} [1 + \cos 2\left(\frac{\theta}{2}\right)]$$

$$= 2 [1 + \cos \theta]$$

$$0 = 2 [1 + \cos \theta]$$

$$\div 2$$

$$0 = 1 + \cos \theta$$

$$\cos \theta = -1$$

$$\theta = \pi$$

$$4 = 2 [1 + \cos \theta]$$

$$\div 2$$

$$2 = 1 + \cos \theta$$

$$\cos \theta = 1$$

$$\theta = 0$$

$$L = 2 \int_0^{\pi} \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2} d\theta$$

$$r = 2 + 2\cos \theta \longrightarrow r^2 = (2 + 2\cos \theta)^2$$

$$= 4 + 8\cos \theta + 4\cos^2 \theta$$

$$\frac{dr}{d\theta} = -2\sin \theta \longrightarrow \left(\frac{dr}{d\theta}\right)^2 = 4\sin^2 \theta$$

$$\Rightarrow r^2 + \left(\frac{dr}{d\theta}\right)^2 = 4 + 8\cos \theta + 4\cos^2 \theta + 4\sin^2 \theta$$

$$= 4 + 8\cos \theta + 4[\cos^2 \theta + \sin^2 \theta]$$

$$= 4 + 8\cos \theta + 4(1) = 8 + 8\cos \theta$$

$$= 8 [1 + \cos \theta]$$

$$L = 2 \int_0^{\pi} \sqrt{8 + 8 \cos \theta} \, d\theta$$

$$= 2 \int_0^{\pi} \sqrt{8(1 + \cos \theta)} \, d\theta$$

$$= 2\sqrt{8} \int_0^{\pi} \sqrt{1 + \cos \theta} \, d\theta$$

مراقة

قانون النصف

$$\cos^2 \theta = \frac{1}{2} [1 + \cos 2\theta]$$

$$2 \cos^2 \theta = 1 + \cos 2\theta$$

$$2 \cos^2 \left(\frac{\theta}{2} \right) = 1 + \cos \theta$$

$$= 2\sqrt{8} \int_0^{\pi} \sqrt{2 \cos^2 \left(\frac{\theta}{2} \right)} \, d\theta$$

$$= 2\sqrt{8} \sqrt{2} \int_0^{\pi} |\cos \frac{\theta}{2}| \, d\theta$$

$$= 2\sqrt{16} \int_0^{\pi} \cos \frac{\theta}{2} \, d\theta$$

$$= 2(4) \left[\sin \frac{\theta}{2} \right]_0^{\pi}$$

$$= 2(4)(2) \left[\sin \frac{\pi}{2} - 0 \right]$$

$$= 16 [1 - 0] = 16$$

من المطلق

$$\frac{\theta}{2} = \frac{\pi}{2}$$

$$\theta = \pi \notin (0, \pi)$$



خطوات اشتقاق الـ Polar



$$1) x = r \cos \theta$$

$$2) y = r \sin \theta$$

إذا كان المطلوب (Equation of (tangent-normal)

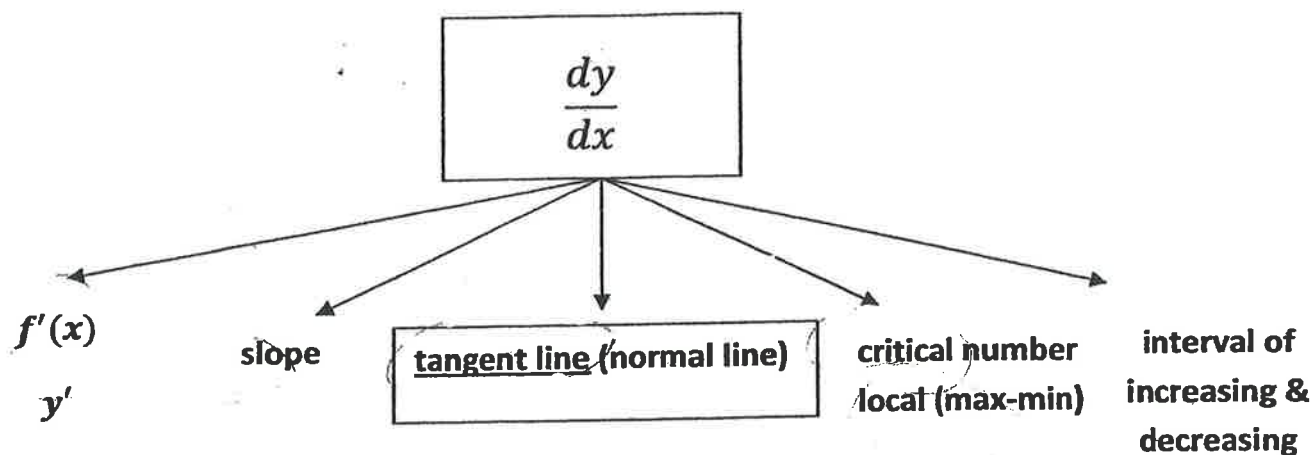
يتم التعويض بالنقطة

$$3) \frac{dx}{d\theta}$$

$$4) \frac{dy}{d\theta}$$

إذا عندي نقطة أعوض فيهم

$$5) \frac{dy}{dx} = - \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}}$$



Equation

$$y - y_0 = m(x - x_0)$$

$$m_{\text{normal}} = \frac{-1}{m_{\text{tangent}}}$$

Horizontal

$$\frac{dy}{dx} = 0$$

أفقار بسط
المشتقة

Vertical

$$\frac{dy}{dx} \text{ undefind}$$

أفقار مقام
المشتقة

Parallel to the line y

$$\frac{dy}{dx} = y'$$

Perpendicular to the line y

$$\frac{dy}{dx} = \frac{-1}{y'}$$

has slope

$$\frac{dy}{dx} = \text{slope}$$

Find the slope of the tangent line to the graph of polar equation $r = e^{3\theta}$ at the point corresponding $\theta = \frac{\pi}{2}$

$$\textcircled{1} \quad x = r \cos \theta \longrightarrow x = e^{3\theta} \cos \theta$$

$$\textcircled{2} \quad y = r \sin \theta \longrightarrow y = e^{3\theta} \sin \theta$$

$$\textcircled{3} \quad \frac{dx}{d\theta} = e^{3\theta} (3) \cos \theta + e^{3\theta} (-\sin \theta)$$

$$= 3e^{3\theta} \cos \theta - e^{3\theta} \sin \theta$$

$$\text{at } \theta = \frac{\pi}{2} \longrightarrow \frac{dx}{d\theta} = 3e^{3\frac{\pi}{2}} (0) - e^{3\frac{\pi}{2}} (1) = -e^{3\frac{\pi}{2}}$$

$$\textcircled{4} \quad \frac{dy}{d\theta} = e^{3\theta} (3) \sin \theta + e^{3\theta} \cos \theta$$

$$\text{at } \theta = \frac{\pi}{2} \longrightarrow \frac{dy}{d\theta} = e^{3\frac{\pi}{2}} (3) (1) + e^{3\frac{\pi}{2}} (0) = 3e^{3\frac{\pi}{2}}$$

$$\textcircled{5} \quad \frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}} = \frac{3e^{3\frac{\pi}{2}}}{-e^{3\frac{\pi}{2}}} = -3$$

$\Rightarrow \therefore$ the slope of the tangent line $= -3$

Extra Find the slope of the normal line

$$= \frac{1}{3}$$

Find an equation in rectangular coordinates of tangent line to the graph of polar equation $r = 4\cos^3\theta$ at the point corresponding $\theta = \frac{\pi}{4}$

$$① \quad x = r \cos \theta \rightarrow x = 4\cos^3\theta \cos \theta = 4\cos^4\theta$$

$$② \quad y = r \sin \theta \rightarrow y = 4\cos^3\theta \sin \theta$$

$$\text{at } \theta = \frac{\pi}{4} \begin{cases} x = 4\left(\frac{1}{\sqrt{2}}\right)^4 = 4 \cdot \frac{1}{4} = 1 \\ y = 4\left(\frac{1}{\sqrt{2}}\right)^3 \cdot \frac{1}{\sqrt{2}} = 4 \cdot \frac{1}{4} = 1 \end{cases}$$

$$\Rightarrow P(1, 1)$$

لان المطلوب
Equation
أعوض بالنقطة



$$③ \quad \frac{dx}{d\theta} = 16\cos^3\theta (-\sin\theta) = -16\cos^3\theta \sin\theta$$

$$\text{at } \theta = \frac{\pi}{4} \rightarrow \frac{dx}{d\theta} = -16\left(\frac{1}{\sqrt{2}}\right)^3 \cdot \frac{1}{\sqrt{2}} = -16 \cdot \frac{1}{4} = -4$$

$$④ \quad \frac{dy}{d\theta} = 12\cos^2\theta (-\sin\theta) \sin\theta + 4\cos^3\theta \cos\theta$$

$$\text{at } \theta = \frac{\pi}{4} \rightarrow \frac{dy}{d\theta} = 12\left(\frac{1}{\sqrt{2}}\right)^2 \left(\frac{1}{\sqrt{2}}\right) \cdot \frac{1}{\sqrt{2}} + 4\left(\frac{1}{\sqrt{2}}\right)^3 \cdot \frac{1}{\sqrt{2}} \\ = 12\left(\frac{1}{4}\right) + 4\left(\frac{1}{4}\right) = -3 + 1 = -2$$

$$⑤ \quad \frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}} = \frac{-2}{-4} = \frac{1}{2}$$

∴ Equation of the tangent line

$$y - y_0 = m(x - x_0)$$

$$y - 1 = \frac{1}{2}(x - 1)$$

Extra:

Find the equation of normal line.

$$y - y_0 = m(x - x_0)$$

$$y - 1 = -2(x - 1)$$

← اقلب تم احزب

Consider the circle $r = \sin\theta$

- Find the intersection point of the circle to the line $\theta = \frac{\pi}{4}$
- Find the points on the circle where the tangent line is parallel to $y = x$

$$y' = 1$$

لايجاد Intersection point

أساوهم مع بعض

أعوض وحدة بالثانية



$$\begin{aligned} \text{a) } r &= \sin\theta & \theta &= \frac{\pi}{4} \\ r &= \sin\frac{\pi}{4} = \frac{1}{\sqrt{2}} & \implies & P(r, \theta) = \left(\frac{1}{\sqrt{2}}, \frac{\pi}{4}\right) \end{aligned}$$

$$\text{b) ① } x = r \cos\theta \longrightarrow x = \sin\theta \cos\theta$$

$$\text{② } y = r \sin\theta \longrightarrow y = \sin\theta \sin\theta = \sin^2\theta$$

$$\begin{aligned} \text{③ } \frac{dx}{d\theta} &= \cos\theta \cdot \cos\theta + \sin\theta (-\sin\theta) \\ &= \cos^2\theta - \sin^2\theta = \cos 2\theta \end{aligned}$$

$$\text{④ } \frac{dy}{d\theta} = 2 \sin\theta \cos\theta = \sin 2\theta$$

$$\text{⑤ } \frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}} = \frac{\sin 2\theta}{\cos 2\theta} = \tan 2\theta$$

$\implies$ see next page plz !

∴ the tangent line of r is parallel to y

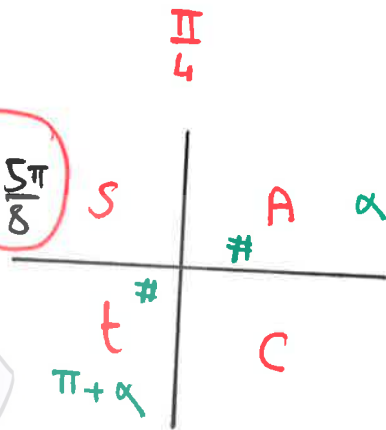
$$\therefore \frac{dy}{dx} = y'$$

$$\tan 2\theta = 1 \longrightarrow$$

شئو $\tan$ الي
تطيني 1

$$2\theta = \frac{\pi}{4} \longrightarrow \theta = \frac{\pi}{8}$$

$$2\theta = \pi + \frac{\pi}{4} \longrightarrow 2\theta = \frac{5\pi}{4} \longrightarrow \theta = \frac{5\pi}{8}$$



$P(x, y)$

$P(r, \theta)$

أختار أكلهما

* يبى أطلع point

أختار أي شئ.

← دام ما حد ربي ←

Sol #1

at $\theta = \frac{\pi}{8}$

$$\longrightarrow r = \sin \frac{\pi}{8} \longrightarrow P\left(\sin \frac{\pi}{8}, \frac{\pi}{8}\right)$$

at $\theta = \frac{5\pi}{8}$

$$\longrightarrow r = \sin \frac{5\pi}{8} \longrightarrow P\left(\sin \frac{5\pi}{8}, \frac{5\pi}{8}\right)$$

Sol #2

at $\theta = \frac{\pi}{8}$

$$x = \sin \frac{\pi}{8} \cos \frac{\pi}{8}$$

$$y = \sin^2 \frac{\pi}{8}$$

$$\longrightarrow P\left(\sin \frac{\pi}{8} \cos \frac{\pi}{8}, \sin^2 \frac{\pi}{8}\right)$$

at $\theta = \frac{5\pi}{8}$

$$x = \sin \frac{5\pi}{8} \cos \frac{5\pi}{8}$$

$$y = \sin^2 \frac{5\pi}{8}$$

$$\longrightarrow P\left(\sin \frac{5\pi}{8} \cos \frac{5\pi}{8}, \sin^2 \frac{5\pi}{8}\right)$$

Let C be the polar curve $r = \sin^3(\theta/3)$, $0 \leq \theta \leq \pi$

a) Find the length of the C .

May 2017

b) Find an equation for the tangent line to C at $\theta = \pi/2$.

$$L = \int_0^{\pi} \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2} d\theta$$

$$r = \sin^3\left(\frac{\theta}{3}\right) \rightarrow r^2 = \left(\sin^3\left(\frac{\theta}{3}\right)\right)^2 = \sin^6\left(\frac{\theta}{3}\right)$$

$$\frac{dr}{d\theta} = 3 \sin^2\left(\frac{\theta}{3}\right) \cos\left(\frac{\theta}{3}\right) \cdot \frac{1}{3} \rightarrow \left(\frac{dr}{d\theta}\right)^2 = \sin^4\left(\frac{\theta}{3}\right) \cos^2\left(\frac{\theta}{3}\right)$$

$$\begin{aligned} \Rightarrow r^2 + \left(\frac{dr}{d\theta}\right)^2 &= \sin^6\left(\frac{\theta}{3}\right) + \sin^4\left(\frac{\theta}{3}\right) \cos^2\left(\frac{\theta}{3}\right) \\ &= \sin^4\left(\frac{\theta}{3}\right) \left[\sin^2\left(\frac{\theta}{3}\right) + \cos^2\left(\frac{\theta}{3}\right) \right] = \sin^4\left(\frac{\theta}{3}\right) \quad \text{①} \end{aligned}$$

$$\therefore L = \int_0^{\pi} \sqrt{\sin^4\left(\frac{\theta}{3}\right)} d\theta$$

$$= \int_0^{\pi} \sin^2\left(\frac{\theta}{3}\right) d\theta$$

$$= \frac{1}{2} \int_0^{\pi} \left[1 - \cos\left(\frac{2\theta}{3}\right) \right] d\theta$$

$$= \frac{1}{2} \left[\theta - \frac{3}{2} \sin\left(\frac{2\theta}{3}\right) \right]_0^{\pi}$$

$$= \frac{1}{2} \left[\pi - \frac{3}{2} \frac{\sqrt{3}}{2} - 0 \right]$$

$$b) \quad x = r \cos \theta \longrightarrow x = \sin^3\left(\frac{\theta}{3}\right) \cos \theta$$

$$y = r \sin \theta \longrightarrow y = \sin^3\left(\frac{\theta}{3}\right) \sin \theta$$

$$\text{at } \theta = \frac{\pi}{2} \begin{cases} x = 0 \\ y = \sin^3\left(\frac{\pi}{6}\right) (1) = \left(\frac{1}{2}\right)^3 = \frac{1}{8} \end{cases} \implies P\left(0, \frac{1}{8}\right)$$

$$\frac{dx}{d\theta} = 3\sin^2\left(\frac{\theta}{3}\right) \cos \theta \left(\frac{\theta}{3}\right) \frac{1}{3} \cos \theta + \sin^3\left(\frac{\theta}{3}\right) (-\sin \theta)$$

$$\text{at } \theta = \frac{\pi}{2} \rightarrow \frac{dx}{d\theta} = 0 + \sin^3\left(\frac{\pi}{6}\right) (-1) = \left(\frac{1}{2}\right)^3 (-1) = -\frac{1}{8}$$

$$\frac{dy}{d\theta} = 3\sin^2\left(\frac{\theta}{3}\right) \cos\left(\frac{\theta}{3}\right) \frac{1}{3} \sin \theta + \sin^3\left(\frac{\theta}{3}\right) \cos \theta$$

$$\begin{aligned} \text{at } \theta = \frac{\pi}{2} \rightarrow \frac{dy}{d\theta} &= 3\sin^2\left(\frac{\pi}{6}\right) \cos\left(\frac{\pi}{6}\right) \frac{1}{3} (1) + 0 \\ &= 3 \left(\frac{1}{2}\right)^2 \frac{\sqrt{3}}{2} \frac{1}{3} = \frac{\sqrt{3}}{8} \end{aligned}$$

$$\frac{dy}{dx} = \frac{\frac{\sqrt{3}}{8}}{-\frac{1}{8}} = -\sqrt{3}$$

∴ The equation of the tangent line :

$$y - y_0 = m(x - x_0)$$

$$y - \frac{1}{8} = -\sqrt{3}(x - 0)$$